

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5. HT	(a)			C = 12 H = 1 N = 14 O = 16 (1) 2C = 24 (1) 3H = 3 (1) Mr = 57 24, 3, 14 and 16 seen – 3 marks		4		4	3	
	(b)			Conversion to 30 000 000 g (1) Number = 30 000 000 / 57 (ecf) (1) = 526 000 (1) Answer of 526 = 2 marks 526 320 = 2 marks 526.32 = 1 mark		3		3	3	
	(c)			Indicative content Reaction pathway: • The reaction pathway shown is for an exothermic reaction. • This means that the difference in PE is released as heat. Bond energies: • During a reaction bonds are broken and formed • More energy released when bonds are formed than when bonds are broken Thermal runaway • Thermal runaway occurred because an increase in temperature arose • which causes an increase in reaction rate • which further increases the temperature • The gas pressure was so great that the valve blew open releasing the toxic gas.		6		6		

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					AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks At least 5 statements taken from all 3 areas <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks At least 3 points from two areas <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Any point from any area <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 5 total		13		13	6	